Hands-on for finding the best R2 value

1. Multiple Linear Regression:
   R2 value is 0.93

2. Support Vector Machine Regression:

		R-value			
S.No	**Regularization parameter**	**Linear**	**Poly**	**RBF**	**Sigmoid**
1	C=10	-0.03	-0.05	-0.05	-0.05
2	C=100	0.11	-0.02	-0.05	-0.03
3	C=500	0.59	0.11	-0.02	0.07
4	C=1000	0.78	0.26	0.01	0.18
5	C=2000	0.87	0.48	0.07	0.39
6	C=3000	0.89	0.63	0.12	0.59

When C=3000 and kernel is Linear, R2 value is 0.89

3. Decision Tree:

S.No	Criterion	Max Features	Splitter	R Value
1	squared_error	None	best	0.90
2	squared_error	None	random	0.91
3	squared_error	sqrt	best	0.21
4	squared_error	sqrt	random	0.26
5	squared_error	Log2	best	0.36
6	squared_error	Log2	random	0.50
7	absolute_error	None	best	0.94
8	absolute_error	None	random	0.91
9	absolute_error	sqrt	best	0.61
10	absolute_error	sqrt	random	0.39
11	absolute_error	Log2	best	0.40
12	absolute_error	Log2	random	0.79

13	Friedman_mse	None	best	0.89
14	Friedman_mse	None	random	-0.11
15	Friedman_mse	sqrt	best	0.32
16	Friedman_mse	sqrt	random	-0.01
17	Friedman_mse	Log2	best	0.86
18	Friedman_mse	Log2	random	-0.42

When Criterion is absolute_error, Max Features is None and Splitter is best, R2 value is 0.94

4. Random Forest Regression

S.No	Criterion	Max Features	N_Estimators	R Value
1	squared_error	None	10	0.93
2	squared_error	None	100	0.94
3	squared_error	sqrt	10	0.57
4	squared_error	sqrt	100	0.82
5	squared_error	Log2	10	0.83
6	squared_error	Log2	100	0.74
7	absolute_error	None	10	0.95
8	absolute_error	None	100	0.93
9	absolute_error	sqrt	10	0.76
10	absolute_error	sqrt	100	0.74
11	absolute_error	Log2	10	0.78
12	absolute_error	Log2	100	0.78
13	friedman_mse	None	10	0.92
14	friedman_mse	None	100	0.94
15	friedman_mse	sqrt	10	0.76
16	friedman_mse	sqrt	100	0.8
17	friedman_mse	Log2	10	0.85
18	friedman_mse	Log2	100	0.76

When Criterion is absolute_error, Max Features is None and N_Estimators is 10, R2 value is 0.95